

OPERATOR QUICK REFERENCE

Low Energy Test Bench

Clutch Endurance Testing Machine

Field	Details
Customer	Valeo Friction Materials India Private Limited
Machine	Low Energy Test Bench — Clutch Endurance Test
Location	R&D Building, R&D Cell, Chennai – 603 204
Document type	Operator Quick Reference (condensed from full Operating & Safety Manual)
Full manual reference	[Insert full manual document number]
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Prepared by	Independent Safety Consultant — B11 LMSS Certified
Classification	Confidential — Authorised Valeo personnel only



This document is a quick reference only. Always refer to the full Operating & Safety Manual for complete procedures, compliance detail, and legal requirements.



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1 Machine Identification Summary

Field	Details
Machine name	Low Energy Test Bench — Clutch Endurance Test
Machine type	Automated endurance test bench
Manufacturer	Not available — retrofit machine (refer OA-001)
Serial / Model number	Not available — no nameplate (refer OA-001)
Control system	New Craft Technology — Job No. 1639, Rev R0, Dec 2025
Location	R&D Building, R&D Cell
Factory	Valeo Friction Materials India Pvt. Ltd., Chennai – 603 204
Main power supply	415V AC, 3-phase, 50 Hz
Control voltage	24V DC
Pneumatic supply	4 to 6 bar (regulated)
Clamping force range	450 to 500 DaN
Test management software	MORPHEE (Windows XP workstation)
Test creation software	MEditor
Safety relay	Pilz PNOZ X3 — dual channel
Drive control	Variable Frequency Drive (VFD)

OPEN ACTION OA-001: Machine nameplate is not available. Serial number, model number, and original manufacturer details must be confirmed by Valeo Friction Materials India Private Limited before this manual is finalised.

2 Control Panel — Button Quick Reference



Button	Label	Function	Mode
Red mushroom head	E-STOP (ARRET)	Emergency stop — twist to release	All modes
White illuminated	RESET	Reset safety relay after E-stop or fault	All modes
Key switch	C	Select Auto / Manual mode — key held by Operator 2	All modes
Push button	F1	Manually rotate clutch (hold-to-run)	Manual only
Push button	F2	Engage / disengage clutch	Manual only
Push button	F3	Clutch bed forward	Manual only
Push button	K3	Clutch bed reverse	Manual only
Push button	F5	Position reset to zero	Manual only
Push button	F7	Position Auto Set	Manual only
Digital display	Clamp Torque	Shows clamping force in DaN — target 450 to 500 DaN	All modes



2.1 Indicator lamp status

Lamp	State	Meaning / Action
Green	ON steady	Control circuit healthy — no action needed
Green	OFF	Control circuit off — check power, isolator, Control ON button
Yellow	ON steady	Test running in auto mode — normal
Yellow	Flashing	Fault or warning — check MORPHEE display
Red	ON steady	E-stop active or safety relay fault — identify cause, reset per Section 5
Red	Flashing	VFD trip or motor fault — refer Section 7 troubleshooting

3 Pre-Start Verification Checklist

Complete before every shift and every test run. Do not start the machine if any item cannot be confirmed.

Item	Check	OK?	Init.
A1	Main panel SSB-02 — LE Bench supply breaker ON	☐	
A2	Machine rotary isolator on control panel — ON	☐	
A3	Green power indicator lamp illuminated (24V DC healthy)	☐	
A4	No MCB or MPCB in control panel in tripped position	☐	
B1	Compressed air supply valve open	☐	
B2	Air pressure regulator reading 4 to 6 bar	☐	
B3	Pneumatic filter bowl — no water or contamination	☐	
C1	Dust collection exhaust motor starter switched ON	☐	
C2	Airflow felt at exhaust outlet	☐	
C3	Air filter — clean, not overdue for replacement	☐	
D1	All fixed guards (G001, G002, G003) in position	☐	
D2	All three machine doors closed and interlock solenoids engaged	☐	
D3	All three E-stop buttons tested — red lamp lights, twist to release	☐	
D4	Pilz PNOZ X3 reset successful — all channel LEDs illuminated	☐	
D5	No persons inside machine enclosure or hazard zone	☐	
E1	MORPHEE running — no startup error, all channels reading	☐	
E2	Test program file selected in MORPHEE	☐	
F1	All required PPE available and serviceable	☐	
F2	Machine area clear — no obstructions in 1 m zone	☐	



DO NOT START if any of these conditions exist:

- Any E-stop button is latched and cannot be released by twisting
- Any door interlock is not engaging
- Air pressure outside 4 to 6 bar range
- Any MCB / MPCB in the tripped position
- MORPHEE displaying a fault or error
- Any guard missing, damaged, or not in position
- Visible smoke, burning smell, oil leak, or other abnormal condition

Pre-start sign-off

Date	Shift	Operator name	Signature

4 Operating Cycle — Step-by-Step Summary

All clutch loading, displacement setting, and manual mode door access must be performed by Operator 2 only.

4.1 Machine starting

- 1 Confirm all pre-start checks (Section 3) are complete and signed off.
- 2 At SSB-02, confirm LE Bench supply breaker is ON.
- 3 Turn machine rotary isolator to ON.
- 4 Twist all three E-stop buttons out fully.
- 5 Press RESET — green lamp illuminates.
- 6 Press Control ON — control circuit energises.
- 7 Confirm dust collection exhaust is running.
- 8 Confirm MORPHEE is running and all channels are healthy.

4.2 Cover plate preparation (at oven — before loading)

⚠ WARNING: Oven operates at 300°C. Use heat-resistant gloves. Allow components to cool before fitting thermocouple.

1. Bolt flywheel (17 mm) + sim washer (7.7 mm) + cover plate using M8 bolts.
2. Place assembly in oven at 300°C for 1.5 hours.
3. Drill 4.8 mm thermocouple holes: centre surface + outer punched rivet of cover plate.
4. Remove from oven using heat-resistant gloves. Allow to cool.
5. Fit thermocouple into both drilled hole locations.

4.3 Clutch assembly loading (Manual mode)

6. Confirm mode key switch (C) is in Manual. Operator 2 holds key.
7. Open machine door. Confirm interlock releases.
8. Place friction disc on drive shaft — align using M8 Aline key bolt and spline.
9. Fit flywheel to adapter using M5 bolt.
10. Fit and tighten cover plate using M8 bolts (cross-pattern).
11. Rotate lead screw wheel — move bed forward until drive shaft connected.
12. Connect thermocouple plug.
13. Tighten M10 slinkey fixing bolt to secure bed.
14. Close machine door. Confirm interlock engages.

4.4 Clutch displacement setting

Target: 450 to 500 DaN on Clamp Torque display.

15. Confirm mode key (C) in Manual.
16. Press F2 — disengage.
17. Press and hold F1 — rotate disc.
18. Press F2 — engage. Check clamp torque display.
 - Below 450 DaN → press F3 (forward). Repeat from Step 2.
 - 450 to 500 DaN → displacement correct. Go to Step 5.
 - Above 500 DaN → press K3 (reverse). Repeat from Step 2.
19. Press F5 — Position Reset to zero.
20. Press K3 — move to -400.
21. Press F5 — second Position Reset.
22. Press F7 — Position Auto Set.
23. Turn mode key (C) to Auto. Machine ready for test launch.

4.5 Test launch

24. In MORPHEE, select the required test program file.
25. Confirm mode key (C) in Auto.
26. Confirm all three doors closed and locked.
27. Press Validation button in MORPHEE. Confirm all conditions met.
28. Press Start. Test begins — MORPHEE records torque, temperature, RPM.

4.6 Test monitoring — each shift

- Check MORPHEE — torque, temperature, RPM within range, no alarms.
- Check dust collection exhaust is running — airflow felt at outlet.
- Check air pressure — 4 to 6 bar.
- Record cycle count in machine log.
- Confirm no one has entered the 1 m machine clearance zone.

4.7 Inspection at 6,000 cycles

⚠ WARNING: Pause test and wait for all motion to stop. Wait for door guard locking unit to release automatically before opening any door.

⚠ CAUTION: High internal temperatures after testing. Wear heat-resistant gloves. Do not touch clutch components with bare hands.

29. Pause or stop test in MORPHEE.
30. Wait for all rotating parts to stop completely.
31. Wait for door guard locking unit to release automatically.
32. Open door. Clean friction disc with compressed air (wear FFP2 respirator).
33. Measure and record wear depth.
34. Assess and record surface bedding percentage — both faces.
35. If within acceptance criteria: reassemble (Section 4.3, Steps 3–9) and relaunch.
36. If worn beyond acceptance: remove clutch (Section 4.8), hand scrapped components to EHS.

4.8 Clutch removal

37. Stop test in MORPHEE. Confirm all motion stopped.
38. Wait for door locking unit to release. Open door.
39. Remove bed fixing bolt.
40. Rotate lead screw wheel — move bed in reverse until drive shaft disconnected.
41. Disconnect thermocouple plug.
42. Remove friction disc, cover plate, and flywheel.
43. Send components to quality inspection station.



4.9 Machine shutdown

44. Stop or pause test in MORPHEE. Wait for motion to stop.
45. Save MORPHEE data files. Close MORPHEE.
46. Mode key (C) to Manual. Remove key. Store securely.
47. Turn rotary isolator to OFF.
48. Switch OFF LE Bench supply at SSB-02.
49. Switch OFF dust collection exhaust starter.
50. Sign off machine log — record cycle count, test status, observations.

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5 Safety Rules and PPE

5.1 Non-negotiable safety rules

- Only authorised, trained personnel may operate or maintain this machine.
- Never enter the 1 m clearance zone while a test is running in automatic mode.
- Never bypass, defeat, or remove any guard, interlock, or safety device.
- Never operate the machine with a known fault.
- Operator 2 must hold the mode key at all times during manual mode.
- Visitors must not enter the machine area without EHS escort.

5.2 PPE requirements

PPE	When required	Who
Safety glasses	All times in machine area	All personnel — mandatory
Hearing protection (ear plugs / muffs)	All test runs	All personnel in area
FFP2 respirator	Friction disc handling, 6,000 cycle inspection, compressed air cleaning	Operator 2
Heat-resistant gloves (min. 300°C)	Oven handling and clutch removal after testing	Operator 2
Safety shoes (steel-toed)	All times in machine area	All personnel



5.3 Emergency stop procedure

E-stop button locations: (1) Operator panel — front of machine. (2) Left side of machine enclosure. (3) Cabin workstation area.

51. Press nearest E-stop immediately. Shout warning to others.
52. Do not touch a person in contact with the machine — call for help.
53. Call EHS Lead and maintenance person (Section 8 — Emergency Contacts).
54. Do not restart until cause is investigated and resolved.

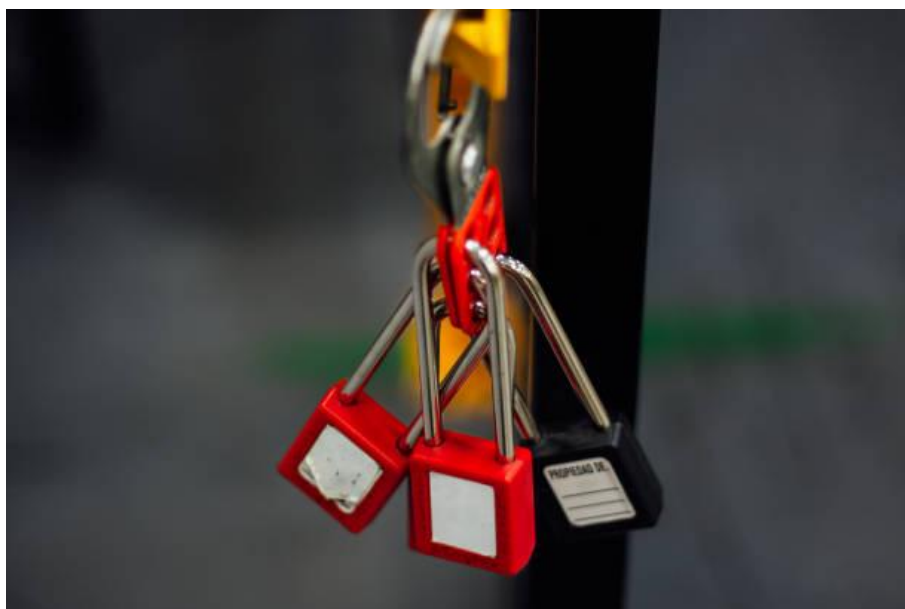
To reset after E-stop:

55. Confirm area is safe and all persons are clear.
56. Identify and resolve the cause.
57. Twist all three E-stop buttons clockwise until they pop out.
58. Press RESET. Press Control ON.
59. Record the E-stop event in the machine log before restarting.



5.4 LOTO — Lockout / Tagout Procedure

LOTO is mandatory before any maintenance activity, guard removal, or physical access inside the machine or control panel. Each person working on the machine must apply their own personal padlock — only that person may remove it.



Energy isolation points

Energy source	Isolation point	Method
Electrical — machine supply	Rotary isolator on machine control panel	Turn to OFF position — padlock + LOTO tag
Electrical — mains supply	SSB-02 panel — LE Bench supply breaker	Switch OFF — padlock + LOTO tag
Pneumatic — compressed air	Pneumatic LOTO valve on supply line to machine	Turn valve to CLOSED — operate exhaust valve to bleed all residual pressure — confirm zero bar on gauge — padlock + LOTO tag

⚠ WARNING: Every person working on the machine must apply their own personal padlock to each isolation point. A padlock may only be removed by the person who applied it. Do not remove another person's padlock under any circumstances.

Tag-out sequence (before starting work)

60. Inform the shift supervisor and EHS Lead that the machine is being isolated for maintenance.
61. Stop the machine using the normal shutdown procedure (Section 4.9) if it is running.
62. At the machine control panel — turn the rotary isolator to OFF. Apply personal padlock and LOTO tag.
63. At SSB-02 panel — switch OFF the LE Bench supply breaker. Apply personal padlock and LOTO tag.
64. At the pneumatic LOTO valve — turn to CLOSED. Operate the exhaust valve to bleed all residual air pressure. Confirm the pressure gauge reads zero.
65. Apply personal padlock and LOTO tag to the pneumatic LOTO valve.
66. LOTO tag must state: your name, date and time, reason for isolation, and your contact number.
67. Attempt to press Control ON — confirm the machine does NOT energise. Zero-energy state confirmed.
68. Record the tag-out in the machine log before beginning any work.

Tag-in sequence (after work is complete)

69. Confirm all maintenance work is complete. All tools removed from machine.
70. Confirm all guards and covers have been replaced and secured.
71. Warn all persons in the area that energy is about to be restored.
72. Remove pneumatic LOTO valve padlock and tag (by the person who applied it only). Open the pneumatic valve. Confirm pressure builds to 4 to 6 bar.
73. Remove SSB-02 breaker padlock and tag. Switch ON the LE Bench supply breaker.
74. Remove rotary isolator padlock and tag. Turn rotary isolator to ON.
75. Complete the full pre-start verification checklist (Section 3) before returning the machine to service.
76. Record the tag-in completion in the machine log — include your name and the time.

5.5 Thermal hazards

- Internal clutch assembly temperatures can reach 300°C during sustained test runs.
- Wear heat-resistant gloves when handling components from the oven or after test completion.
- Wait for the door guard locking unit to release automatically — this confirms it is safe to open the door.
- Do not place flammable materials within 1 m of the machine during testing.



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5.6 Dust hazard

⚠ CAUTION: Friction disc dust contains glass fibre and metal. This is a respiratory hazard. Always wear FFP2 respirator when handling friction discs or cleaning with compressed air.

- Dust collection exhaust must run at all times during testing.
- Dispose of used filters and scrap friction discs through the Valeo EHS team — not general waste.
- Wash hands after all friction disc handling.



6 Alarms and Troubleshooting — Quick Reference

Fault	Symptom	Likely cause	Action
Control circuit does not energise	Green lamp off after Control ON	Isolator OFF, MCB tripped, E-stop latched, or safety relay not reset	Check isolator ON. Check all MCBs up. Release all E-stops. Press RESET then Control ON.
Safety relay not resetting	Red lamp stays ON after E-stop release and reset	E-stop not fully released, door open, or PNOZ X3 fault	Confirm all E-stops twisted out. Confirm all doors closed and locked. Check PNOZ X3 LEDs — call maintenance if not resolved.
VFD trip	Red lamp flashing, motor stops	Overcurrent, overtemp, overvoltage, or phase loss	Note VFD fault code. Allow 5 min cooling. Check motor not jammed. If persistent — call specialist.
Door interlock not engaging	Machine will not start in auto mode	Door not fully closed, or solenoid fault	Confirm door is physically closed. If solenoid still fails — call maintenance.
Air pressure fault	MORPHEE fault display, machine stops	Pressure below 4 bar or above 6 bar, valve closed, or regulator fault	Check supply valve is open. Adjust regulator to 4 to 6 bar. If cannot maintain — call maintenance.
Torque reading zero or erratic	MORPHEE shows 0 DaN or erratic torque	Load cell disconnected or calibration drift	Stop test. Check connections. Perform torque calibration check. Call maintenance if unresolved.
Temperature reading zero or max	MORPHEE temperature channel error	Thermocouple disconnected or damaged	Stop test. Check thermocouple plug is connected. Call maintenance if still incorrect.
MORPHEE will not start	Black screen or error on startup	Software or NI card communication fault	Record exact error message. Do not reinstall software. Call New Craft Technology specialist.

7 Maintenance Schedule Summary

OPEN ACTION OA-005: Maintenance intervals below are based on standard industrial practice. Valeo Friction Materials India Private Limited must formally confirm all intervals before this document is finalised.

Activity	Frequency	Who	LOTO?
Pre-start verification checklist (Section 3)	Every shift	Operator 1 or 2	No
Air filter inspection — visual	Weekly	Operator 1 or 2	No
Pneumatic filter bowl drain	Weekly	Operator 1 or 2	No
General housekeeping — external clean	Weekly	Operator 1 or 2	No
Air filter replacement	Monthly or as required	Operator 2 + Maintenance	Yes
Torque calibration check (NABL vendor)	Every 6 months	Maintenance + Calibration vendor	No
Temperature calibration check (NABL vendor)	Every 6 months	Maintenance + Calibration vendor	No
RPM calibration check (NABL vendor)	Every 6 months	Maintenance + Calibration vendor	No
Dust collection duct and fan inspection	Every 6 months	Maintenance person	Yes
Electrical cabinet inspection (specialist)	Every 6 months	New Craft Technology	Yes
Pilz PNOZ X3 full function test (specialist)	Every 12 months	New Craft Technology	Yes
Pneumatic cylinder and actuator inspection	Every 12 months	Maintenance / New Craft Technology	Yes

7.1 Air filter replacement summary

⚠ CAUTION: Complete LOTO before accessing filter housing. Used filters contain glass fibre and metal dust — dispose through Valeo EHS team.

77. Complete LOTO procedure (full procedure in Operating & Safety Manual, Section 7.5).
78. Open filter housing access panel.
79. Remove used filter into a sealed plastic bag — do not shake or blow out.
80. Insert new filter in correct orientation (check directional arrow on filter frame).
81. Close and secure access panel.
82. Complete LOTO tag-in. Start machine. Confirm exhaust running correctly.
83. Record replacement in machine log.



7.2 Calibration summary

Parameter	Method (summary)	Acceptance criteria
Torque	1 metre arm bar + calibration masses: 5 / 15 / 25 / 35 / 45 kg. Compare MORPHEE reading vs calculated DaN reference.	Within tolerance at all mass steps — confirm with NABL calibration vendor
Temperature	Fluke reference thermometer. Reference points: 0 / 40 / 80 / 100 / 200 / 300°C.	MORPHEE reading within $\pm 2^{\circ}\text{C}$ of reference thermometer at each point
RPM	Calibrated tachometer. Setpoints: 500 / 1000 / 1500 RPM. Enter manual RPM in CVM field in MORPHEE.	MORPHEE reading within ± 10 RPM of tachometer at each setpoint



8 Emergency Contacts

Role	Name and contact number
EHS Lead — Valeo Friction Materials India Pvt. Ltd.	[Insert name and phone number]
R&D Deputy Manager	[Insert name and phone number]
Maintenance Person (in-house)	[Insert name and phone number]
Third-party Specialist — New Craft Technology (Job No. 1639)	[Insert name and phone number]
Plant Security	[Insert phone number]
Ambulance / Emergency services	108 (Tamil Nadu)
Fire services	101
Local police	100



9 Open Actions Register — Summary

All open actions below must be formally closed by Valeo Friction Materials India Private Limited before this manual is considered complete. Refer to the full Operating & Safety Manual for complete descriptions.

OA No.	Description (summary)	Responsible party	Status
OA-001	Machine nameplate data (serial, model, manufacturer) not available — retrofit machine.	Valeo	Open
OA-002	Drive motor and gear motor electrical ratings not confirmed — must be verified from nameplates.	Valeo	Open
OA-003	Busbar acrylic guard G004 — fabrication and installation not confirmed.	Valeo / New Craft Technology	Open
OA-004	Overvoltage surge protection device — installation not confirmed.	Valeo	Open
OA-005	Maintenance frequency intervals not formally confirmed by customer.	Valeo	Open
OA-006	Commissioning protocol not formally documented or approved by customer.	Valeo / New Craft Technology	Open
OA-007	Document number not assigned — confirm per Valeo document control procedure.	Valeo	Open

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